Quantum Algorithms for Optimization of Constraint Problem
(Traffic Network Partitioning using QAOA)

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Introduction

Many real-world optimization problems, especially on graphs, are NP-hard and cannot be solved efficiently using classical brute-force or exhaustive search. Traffic-network partitioning is a fundamental optimization problem with applications in congestion management and intelligent transportation systems. We formulate the problem as a **weighted Max-Cut** on a graph, where nodes represent intersections and edge weights encode traffic intensity. The Quantum Approximate Optimization Algorithm (QAOA) is employed to approximate the optimal partition. Simulation results on a weighted toy network show that QAOA effectively separates high-intersection regions and outperforms random partitions.

Objectives

- Formulate traffic-network partitioning as a weighted Max-Cut problem.
- Design and implement QAOA to obtain high-quality graph partitions.
- Analyze the effect of circuit depth and parameter optimization on performance.
- Compare QAOA results with random baseline partitions.

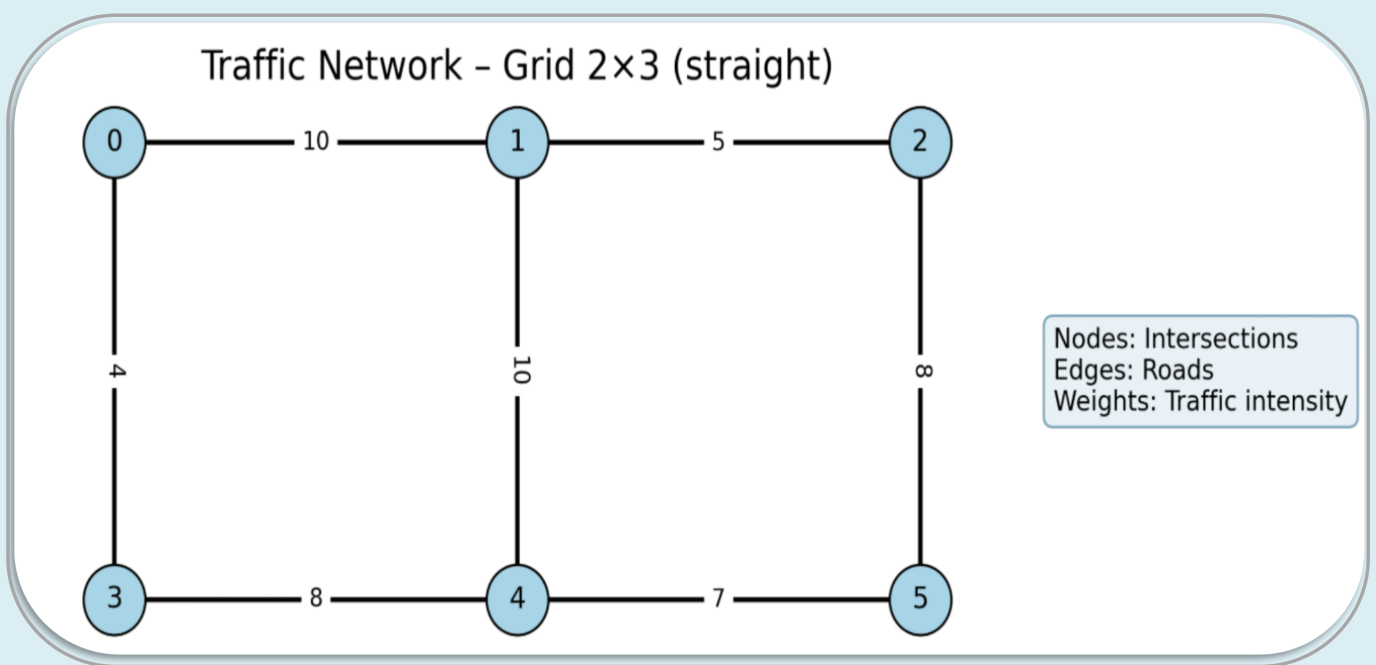
Problem Setup & Methodology

1. Problem Formulation: Weighted Max-Cut

- A traffic network is modeled as a weighted graph G(V,E)
- **Objective:** Partitioning V into two sets maximize the total weight of edges across the cut.
- Weighted Max-Cut Hamiltonian with balance partition constraint

$$\boxed{H_C = \underbrace{\sum_{(i,j)\in E} w_{ij}\,\frac{1 - Z_i Z_j}{2}}_{\text{Weighted Max-Cut}} \;-\; \underbrace{\lambda\left(\sum_i x_i - \frac{n}{2}\right)^2}_{\text{Balance Constraint}}}$$

where Z_i, Z_j are Pauli-Z operators encoding binary partition assignment



2. QAOA Framework

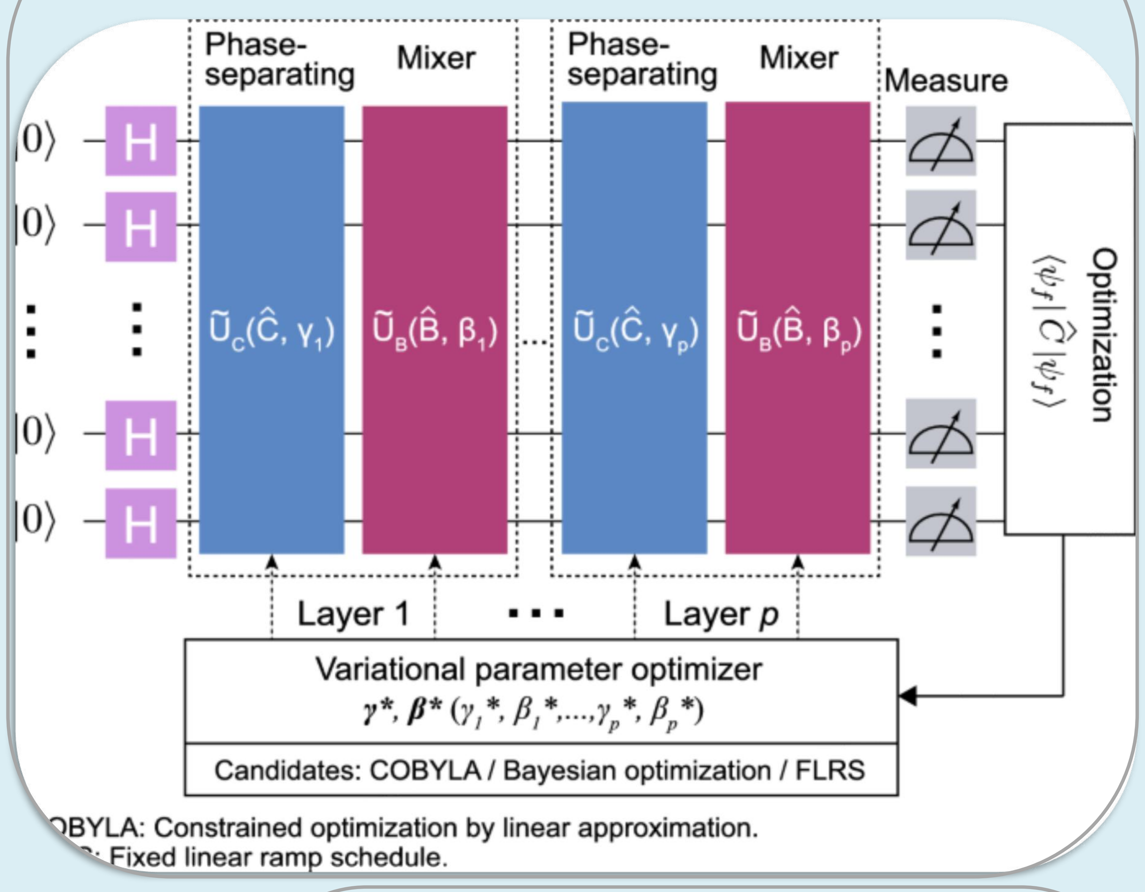
A p-depth QAOA alternates between:

- Cost Hamiltonian Evolution $U_C(\gamma) = e^{(-i\gamma H_C)}$
- Mixer Hamiltonian Evolution $U_M(\gamma) = e^{(-i\gamma H_M)}$

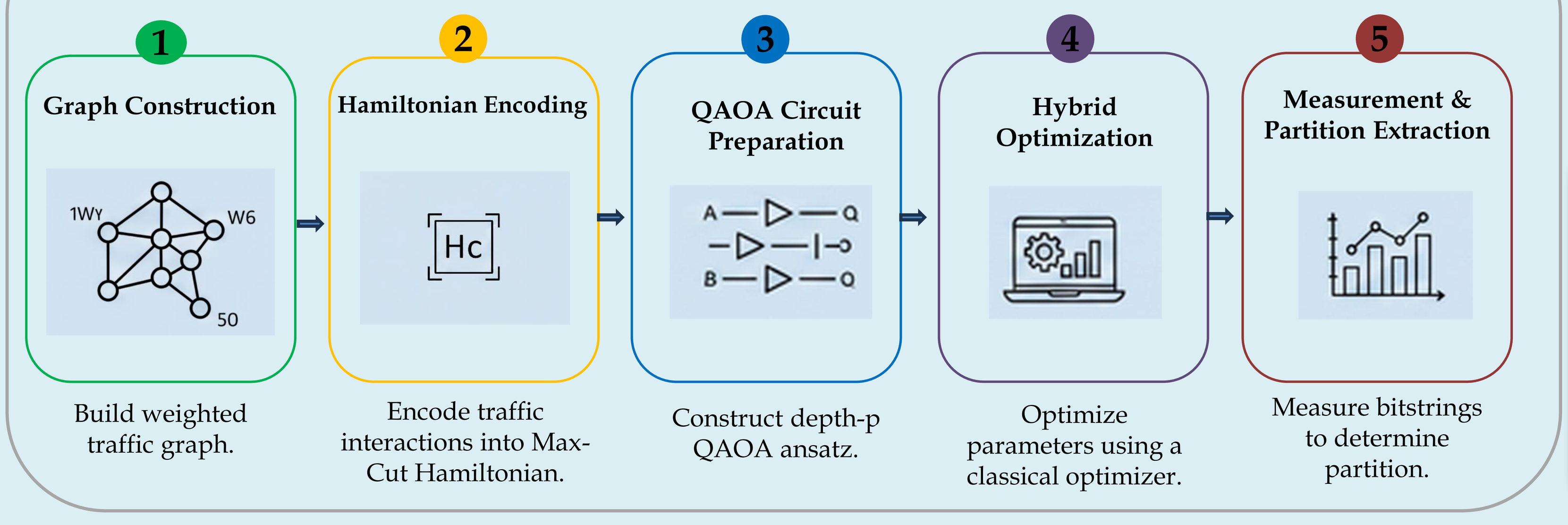
with mixer $H_M = \sum X_i$

The parameters $\vec{\theta} = (\gamma_1, \beta_1, \ldots, \gamma_p, \beta_p)$ are optimize to maximize $< H_C >$.

Quantum Circuit (p layers)

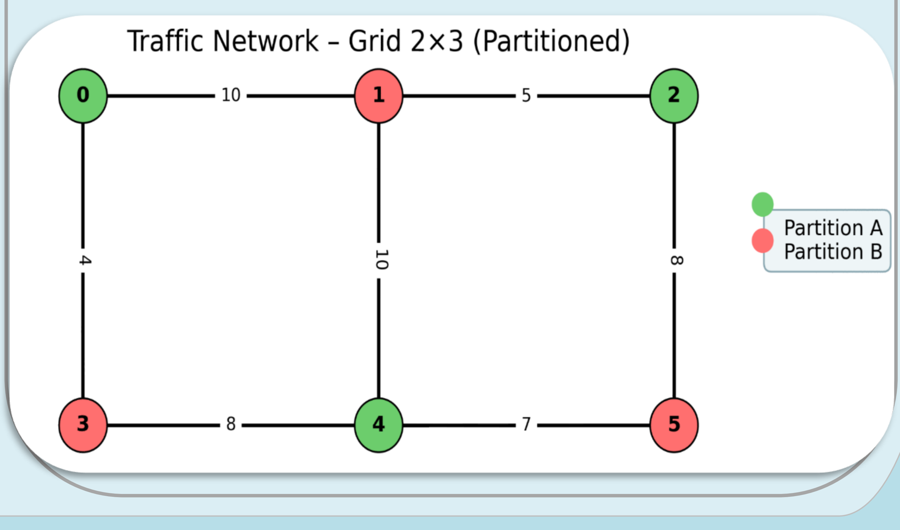


3. Workflow

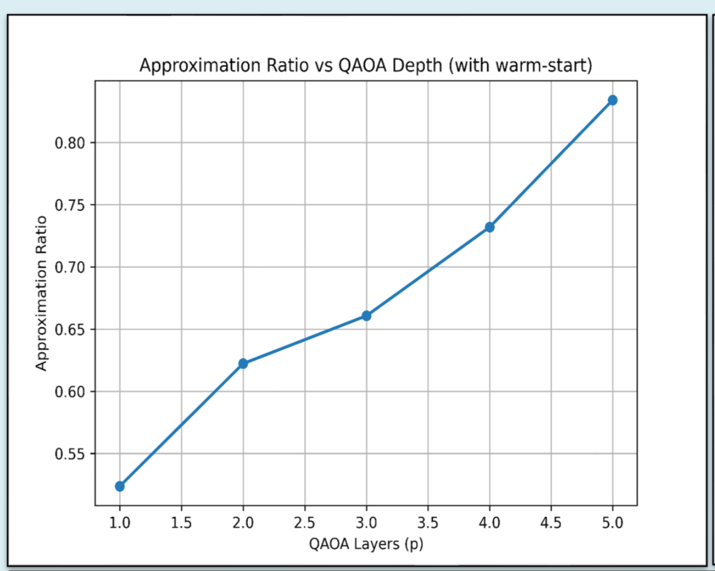


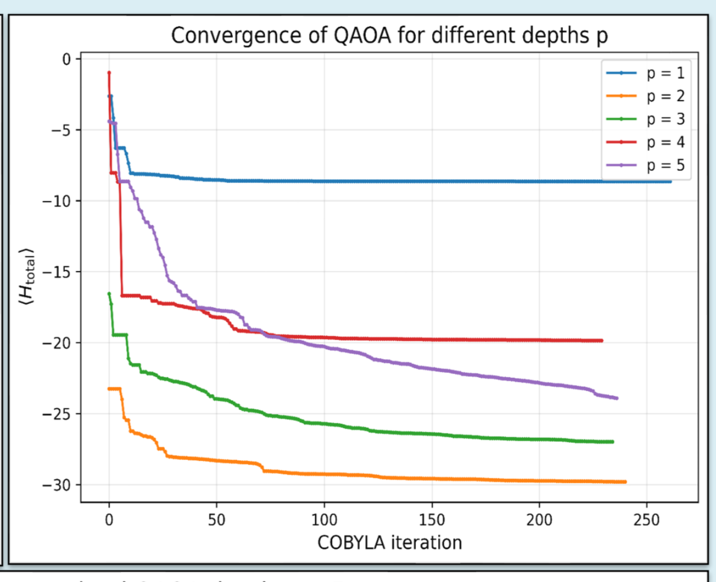
4. Partition Interpretation

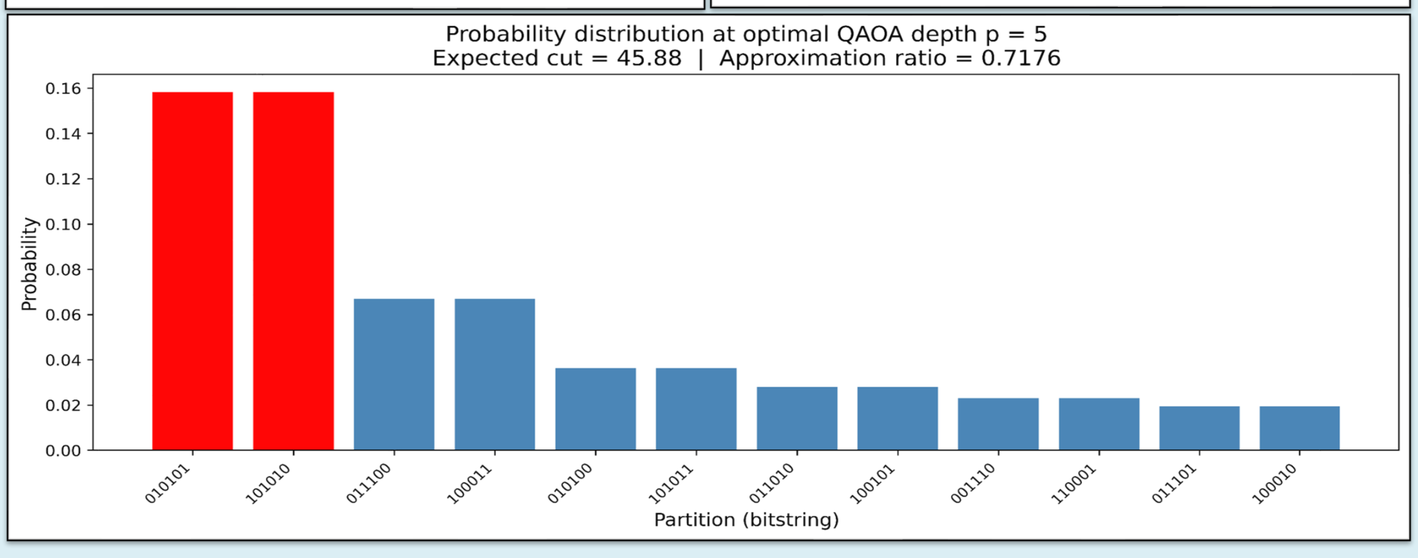
- Bit value 0 → Partition A
- Bit value 1 → Partition B



Results & Conclusion







Conclusion

- QAOA effectively models the traffic-network partitioning problem.
- Weighted Max-Cut formulation captures high-traffic separation.
- Toy grid results confirm correct partition and algorithm validity.
- Demonstrates potential of hybrid quantum–classical optimization.

Future Direction

- Scale to larger, real-world traffic networks.
- Test custom constraint-preserving mixers.
- Incorporate warm-start or adaptive QAOA methods.
- Benchmark against classical heuristics. Implement on real quantum hardware.

References

1. E. Farhi, J. Goldstone, and S. Gutmann, "A quantum approximate optimization algorithm," arXyh biv preprint, vol. arXiv:1411.4028, 2014. [quant-ph].
2. E. Bae and S. Lee, "Recursive QAOA outperforms the original QAOA for the MAX-CUT problem on complete graphs," arXiv preprint, 2023.

Acknowledgement

I sincerely thank the National Centre for Physics (NCP) and PIEAS for providing facilities, computational resources, and a supportive research environment. I am also grateful to my colleague, Sara Nargis, for invaluable collaboration and insightful discussions throughout this work.

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